LECTURE L27-28 · MODULE 4 · CO4

Dimensionality reduction

Week 14 | Slides: `DATA209_Module_4_Feature_Engineering_and_Structure.pptx`, pages 21–28 | Laboratory: **P27-28**

TOPICS COVERED

- PCA — interpretation focused
- LDA — the classification view
- t-SNE and UMAP for visualisation only
- Interpreting components; multicollinearity

Selection versus reduction

Selection keeps a subset of the original columns; the output stays interpretable and the information in dropped columns is discarded. **Reduction** builds new columns as combinations of the originals; interpretation is harder but information from every column is retained.

Prefer selection when interpretability is the deliverable, reduction when variables are collinear or you need a 2-D view. Say which you did and why.

PCA in five steps

1. **Standardise** — centre and scale. Skipping this makes PCA measure units, not structure.
2. **Covariance** — compute the covariance matrix of the standardised features.
3. **Decompose** — find eigenvectors (directions of variance) and eigenvalues (how much lies along each).
4. **Order and cut** — sort by eigenvalue; keep as many as your variance target requires.
5. **Project** — transform onto the retained components, each orthogonal to all the others.

PCA is **unsupervised**: it never looks at the target, so the direction of greatest variance need not be the useful direction.

Reading the output

Retention rules: a cumulative variance target (commonly 80–95%), the scree elbow, or the Kaiser rule (eigenvalue > 1). State which you used.

On the transformed shoppers features, PC1 explains 47.9% and four components reach 79.7% — a meaningful compression. On standardised `pima.csv`, the profile is 28.5 / 18.7 / 14.3 / 11.5 …, reaching 80% only at the fifth component. **A flat scree plot is a finding**, not a failure: it means the variables were already fairly independent.

Interpreting components

Read the **loadings** — the weight of each original variable in the component. A component with large weights on Glucose, Insulin and BMI is a "metabolic load" axis; name it. Signs matter: opposite signs mean the component contrasts two groups of variables.

> A component you cannot name is a component you should not present to a domain audience.

Cautions: scale first; PCA is linear only; it is not appropriate for one-hot encoded categories, whose binary geometry breaks its assumptions; outliers distort components; and it ignores the target.

Multicollinearity detection with VIF

Reduction and multicollinearity are two responses to the same problem, so check for it before you reduce. The **variance inflation factor** for a predictor is 1 / (1 - R^2) from regressing that predictor on all the others: above 5 it is worth investigating, above 10 the column is largely reconstructable from the rest. (Introduced in L11-12; applied here as a screening step.)

Three responses, in increasing order of cost to interpretability:

1. **Drop** one of the pair — cheapest, and defensible when the two measure the same thing.
2. **Combine** them into a single constructed feature — keeps the information, keeps a name.
3. **Reduce** with PCA — keeps all the information and produces orthogonal components by

construction, so multicollinearity is eliminated rather than managed. The cost is that the components need naming before anyone can act on them.

Note the asymmetry: PCA *removes* multicollinearity but does not tell you which original variable was redundant. If the deliverable is a recommendation about which measurements to keep collecting, use VIF and selection instead.

Alternatives

METHOD	SUPERVISED	OPTIMISES	USE FOR
PCA	no	maximum retained variance	compression, decorrelation, interpretation
LDA	yes	maximum class separation	reduction with a label; at most k−1 components
t-SNE	no	local neighbourhood preservation	2-D visualisation only
UMAP	no	local and some global structure	2-D visualisation; faster, more stable

t-SNE and UMAP traps: cluster sizes and between-cluster distances are not meaningful, and results change with the seed and with perplexity. Never use their output as model input.

Finally: PCA belongs **inside** the pipeline, refitted on each training fold. Fitting it on the whole dataset before cross-validation leaks test information.

KEY TERMS

`Principal component analysis` `Eigenvector` `Eigenvalue` `Explained variance ratio` `Scree plot` `Loading` `Kaiser rule` `LDA` `t-SNE` `UMAP`

COMMON ERRORS TO AVOID

- Running PCA on unstandardised features
- Applying PCA to one-hot encoded categorical columns
- Reading cluster sizes or distances off a t-SNE plot
- Fitting PCA on the full dataset before cross-validation

READINGS — ALL FREE TO ACCESS

1. scikit-learn — decomposition and PCA
   `https://scikit-learn.org/stable/modules/decomposition.html`
2. Stanford CS229 — lecture notes
   `https://cs229.stanford.edu/`
3. James et al. — Introduction to Statistical Learning, Ch. 12
   `https://www.statlearning.com/`

TEST YOURSELF

1. Why must features be standardised before PCA?
2. PC1 explains 12% of variance. What does that tell you about the data?
3. Why is t-SNE unsuitable as a preprocessing step for a classifier?
4. How do you turn a loading vector into something a domain expert can use?